Federal Bureau of Investigation (FBI)
Attachment A – FBI Enterprise BAA Problem Sets

1. Leverage Artificial Intelligence (AI) and its Sub-Fields

1.1. **Content Extraction and Triage:** The FBI seeks increased understanding and enhanced ability to apply AI techniques to thematic content extraction and triage, in order to identify common themes and patterns and more efficiently capture, deconstruct, analyze, and reconstruct data from unstructured sources and formats. Certain sources and formats may also require additional pre-processing tools, such as optical character recognition (OCR).

1.2. **All-Source Data Management:** The FBI seeks increased understanding and enhanced ability to apply AI techniques to conduct all-source data collection, research, information monitoring, automated reporting, and database development. Solutions must be able to:

- Streamline and accelerate certain investigation and intelligence workflows
- Compress investigation and intelligence production timelines
- Integrate multiple data sets and formats
- Identify data relationships, connections, correlations, and associations
- Process data and discern relevance with respect to a given topic
- Facilitate information sharing
- Prepare data for anticipatory analysis

Solutions must also act on large amounts of unstructured data in various formats, as well as potentially accommodate dynamic data flows. Solutions must also streamline and accelerate time-consuming, labor-intensive tasks, such as analyzing unstructured audio, images/visualizations, and video.

The FBI also seeks to understand industry capabilities for applying AI techniques, approaches, and solutions to thematic data management and data transformation in a manner that allows a semi-automated approach to the extract-transform-load (ETL) process for collected data. Specific ETL functions include the ability to select, focus, simplify, tag, and transform data into human and/or machine-interpretable form for analysis or action.

1.3. **Workflow Automation:** The FBI seeks increased understanding and enhanced ability to apply AI techniques to automate law enforcement, records management, and intelligence production, planning, and process workflows in order to reduce manual intervention.

1.4. **Actionable Analysis and Alerting:** The FBI seeks to apply advanced analysis algorithms within a big data framework. Specifically, the FBI seeks semi-automated tools that conduct analysis of open-source data, sensor data (e.g., intelligence, human, technology, etc.), finished intelligence, financial intelligence, and other forms of intelligence and investigation reporting to locate trends, create alerts, and recommend actions based on sensor feeds. Solutions must provide modeling techniques, as well as analyze and visually display analytics.

1.5. **Semi-Autonomous Multi-Sensor Fusion:** The FBI seeks to leverage AI-enabled sensor processors to manage and fuse multiple sensors of the same or different phenomenology, recognize and respond to signature sources by type, and identify anomalies. The AI-enabled sensor network will subsequently identify and respond to recognized threat types and threat activity, based on these sensory feeds. Sensory feeds may incorporate analog and digital feeds from acoustic, seismic, magnetic, electro-optic/infrared, density/pressure, electromagnetic, radio frequency, hyperspectral, and other domain signatures, correlated in both space and time.

1.6. **Business Operations:** The FBI seeks increased understanding and enhanced ability to apply AI techniques to business operations, including acquisition management, financial analysis, portfolio prioritization and optimization, business analytics, risk management, records management, resource conservation, and business decision support. Solutions must enhance the FBI's ability to streamline and gain insight through predictive analysis of business financial operations.

1.7. **Data Science:** The FBI seeks increased understanding and enhanced ability to apply AI techniques in data science workflows, including capabilities that support and streamline the processing of text, audio, image, and video data (including Natural Language Processing), netflow data, and other kinds of complex multi-dimensional data.

1.8. **Knowledge Management:** The FBI seeks increased understanding and enhanced ability to apply AI techniques to process finished law enforcement and intelligence products and create a centralized body of organized knowledge for use in exploring topics and concepts. The body of knowledge must return answers to users that describe the connections between data, up to the point in time when they requested the information. Solutions must return non-duplicated and deconflicted data sources to provide analysts with a comprehensive end-product with accurate sourcing.

1.9. **Open-Source Information Gathering:** The FBI seeks increased understanding and enhanced ability to apply AI techniques to semi-autonomous collection of all forms of open-source information in accordance with FBI authorities. Solutions must include the ability to combine, compare, and analyze classified and open-source material, as well as to cross-verify information obtained from different sources.

1.10. **Correspondence Management from Multiple Sources:** The FBI seeks increased understanding and enhanced ability to apply AI techniques to process incoming taskings, public tips, comments, and complaints, etc., from multiple sources, particularly email and online, web-based submissions. Solutions must read submission flows in order to discover, track, and route these types of correspondence through proper channels.

1.11. **Dynamic Threat Analysis:** The FBI seeks increased understanding and enhanced ability to apply AI techniques to dynamic threat analysis modeling. Solutions must provide an automated system that allows analysts to develop weighted analytical threat models leveraging quantified adversary capabilities, doctrine, influence, historical behavior, and stated intent.

1.12. **Human Resource (HR) Recruiting:** The FBI seeks increased understanding and enhanced ability to apply AI techniques to HR recruiting. Solutions must provide a capability that correlates structured and unstructured data sources to identify, recommend, and rank individuals to fill unique billets by leveraging personal biographies, expertise profiles, and available online data. Solutions must also include measures that prevent and/or mitigate bias, identify any biases that are not easily captured, and explain how the solution reached a particular conclusion.

1.13. **Signature Identification:** The FBI seeks increased understanding and enhanced ability to apply AI techniques to identify signatures from various sensor sources to aid in exploitation, analysis, and production. Signature identification supports wireless, radio, acoustic, and chemical and biological Measurement and Signature Intelligence (MASINT) capabilities.

1.14. **Cybersecurity:** The FBI seeks increased understanding and enhanced ability to apply AI techniques to cybersecurity challenges. Solutions must dynamically detect anomalies, risks, and threats, and alert defenders on a larger scale and at a faster pace than what can currently be accomplished using traditional "collect, process, and analyze" methodologies.

1.15. **Artificial Intelligence (AI) Governance:** The FBI seeks innovative approaches, methods, research, and technology to manage Artificial Intelligence (AI) systems and ensure that they operate in accordance with law and policy, to include ethical constraints such as those described in *Promoting the Use of Trustworthy Artificial Intelligence in the Federal Government* (EO 13960) the ODNI *Artificial Intelligence Ethics Framework for the Intelligence Community.*

2. Enhance Counterintelligence and Security

2.1 **Counterintelligence (CI) Capability:** The FBI seeks a new/enhanced counterintelligence capability to improve the timeliness, quality, quantity, and relevance of information gathered in support of CI consumers, in order to exploit data collected by CI assets and to better yield critical information.

2.2 **Personnel Assessment and Evaluation:** In accordance with federal mandates, the FBI seeks to integrate existing FBI insider threat capabilities, technologies, and data sources to enhance the way personnel are assessed for continued eligibility and access to classified information. The FBI will implement an enhanced periodic reinvestigation program by leveraging continuous monitoring and evaluation tools which are intrinsically linked to security, counterintelligence, and insider threat efforts.

Solutions must include a strategy to address multiple aspects of personnel security (including automating processes in order to provide real-time assessment of workforce risks); monitor data sources to flag pertinent information (from an adjudicative and insider threat perspective); and integrate publicly available electronic information (e.g. social media) with traditional background information.

2.3 **Cyber Behavior:** The FBI seeks to develop a system to manage and assess cyberspace behaviors and human persona activities. In particular, the FBI seeks models and algorithms to identify rogue and illicit human persona behaviors engaged in cyberspace activities during authorized and approved lawful surveillance.

2.4 **Foreign Contacts:** The FBI seeks to develop a network tool to search multiple databases for identity data and relevant intelligence reports for identified foreign contacts, and subsequently export this data in a single medium to support analytical reviews.

3. Mission-Enhancing Science and Technology

3.1 **Digital Exploitation:** The FBI seeks to improve next generation capabilities and platforms in order to advance the FBI enterprise. Solutions must include techniques developed for commercial and/or non-intelligence applications that are applied in an innovative manner to address current intelligence gaps.

3.1.1 **Document and Media Acquisition:** The FBI seeks to leverage advances in the field of document and media acquisition, to include data recovery from damaged sources, non-traditional media source acquisition, rapid data acquisition, and methods of counter-destruction of document and media sources.

3.1.2 **Human Language Technologies:** The FBI seeks to leverage advances in the field of human language technology across any medium and modality – including, but not limited to – language and dialect identification, machine translation, categorization, and authorship analytics.

3.1.3 **Computer Vision:** The FBI seeks to leverage advances in the field of computer vision – including, but not limited to – object detection, object recognition, digital biometrics, scene identification, content characterization, and content change detection.

3.1.4 **Digital Signals Processing:** The FBI seeks to leverage advances in the field of digital signals processing – including, but not limited to – signature generation and identification, non-visible/audible spectrum analysis, event characterization and storyboarding, and novel methods for content similarity.

3.1.5 **Data, Process, and User Artifact Analysis:** The FBI seeks to leverage advances in the field of data, process, and user artifact analysis via digital forensics methods – including, but not limited to – temporal/geospatial reconstruction, network activity, application analysis, synthetic media detection and analysis, social media analysis, social network analysis, and digital persona analysis.

3.1.6 **Document and Media Exploitation (DOMEX):** The FBI seeks to leverage advances in DOMEX extraction and cross-domain analytics to improve data discovery and dissemination using cloud AI/ML-based approaches to move “dirty” forensic images to a separate cloud-based analytical environment.

3.2 **Directed Energy Weapons (DEW):** The FBI seeks to understand advances in the collection and analysis of DEW. Solutions must address capabilities and methodologies used to understand the increasing threat, use, and emergence of DEW capabilities, including lasers, electromagnetic pulse (EMP), high powered microwave (HPM), and energetic ballistic weapons. Solutions must also focus on improving technical collection, processing, exploitation, and signature.

3.3 **Biometrics:** The FBI seeks to leverage new and emerging technologies that can be developed, modified, and/or rapidly transitioned to operational biometrics (i.e., human DNA, fingerprint) targeting, collection and exploitation environments, consistent with the law and privacy and civil liberties protections.

3.4 **Weapons of Mass Destruction:** The FBI seeks to understand new and emerging technologies that can be developed, modified, and/or rapidly transitioned to operational Chemical, Biological, Radiological and Nuclear (CBRN) targeting, collection and exploitation environments.

3.5 **Electro-Optical Detection:** The FBI seeks to improve the detection limits of weak emissions associated with Visible and Near-infrared (VNIR), Near-infrared (NIR), Short-wave infrared (SWIR), and Medium-wave Infrared (MWIR). Solutions must substantially improve on the trade space of sensitivity, size, weight, and power (SWAP), as well as noise characteristics currently available in commercial-off-the-shelf (COTS) and government-off-the-shelf (GOTS) products.

Technologies that reduce the need for cooling are of particular interest. Technologies may include sensor technologies themselves, or support technologies, such as innovative algorithmic approaches or nontraditional optical concepts that better leverage existing sensor technologies. The FBI prefers simultaneous multi-spectral capabilities; however, single-wavelength solutions will be considered.

3.6 **Pulsed Power Detection:** The FBI seeks innovative approaches, methods, and technologies that enable improved detection and characterization of pulsed power sources used in weapons, radars, electromagnetic launch systems, and high energy scientific equipment.

3.7 **Radio Frequency Detection:** The FBI seeks innovative approaches, methods, and technologies to improve the detection and characterization of generic radiofrequency (RF) emissions across all bands. Solutions must substantially improve on the trade space of sensitivity, bandwidth, resolution, size, weight, and power (SWAP), as well as noise characteristics currently available in commercial-off-the-shelf (COTS) and government-of-the-shelf (GOTS) solutions.

4. Improve Mission Support Capabilities

4.1 **Information Technology (IT):** The FBI seeks innovative approaches, methods, and technologies to provide IT systems and services to the law enforcement and intelligence communities to comply with national policy and federal acquisition authorities. Machine Learning tools can improve infrastructure availability by assessing the most frequent incident types, and subsequent actions taken to resolve these issues. This information can then be used to create scripts that automatically run when specific conditions are met. This approach will reduce the need for human intervention and should improve system performance and availability.

4.1.1 **IT Governance:** The FBI seeks innovative approaches, methods, and technologies to ensure high levels of satisfaction for consumers of enterprise IT services and products. Solutions must provide front-line support and accountability for customer and mission outcomes through consistent and deliberate interactions, and enable timely and effective decision-making, strategic alignment of IT to mission, measurable outcomes, and customer value. Solutions must consider which constructs will be included in a formal governance framework that guides FBI strategic planning, enterprise architecture, portfolio management, and investment management activities.

4.1.2 **Data Management and Analytics:** The FBI seeks innovative approaches, methods, and technologies to provide data integration engineering services to the FBI enterprise by ingesting, exposing, tagging, and deploying mission-effective, standardized data services and common analytical capabilities.

4.1.3 **Infrastructure and Cloud Management:** The FBI seeks innovative approaches, methods, and technologies to provide data integration engineering services to the FBI enterprise by ingesting, exposing, tagging, and deploying mission-effective, standardized data services and common analytical capabilities.

4.1.4 **Information Assurance and Cyber Defense:** The FBI seeks innovative approaches, methods, and technologies to improve information assurance (IA) and computer network defense (CND) services for the FBI enterprise and the broader law enforcement and intelligence communities. Solutions must aid in providing technical oversight and guidance at all levels on information technology fielding decisions, secure communications, and network security issues.

4.1.5 **Operations Support:** The FBI seeks innovative approaches, methods, and technologies to facilitate vehicle management, sustainment support to field elements, expeditionary support in an appropriately attributable manner to platforms performing contingency and joint operations, and solutions to provide logistics support and physical infrastructure worldwide.

4.1.6 **Strategic Information and Operations (SIOC):** The FBI seeks innovative approaches, methods, and technologies that enhance capabilities to provide FBI leadership, law enforcement, and intelligence community partners with 24/7 situational awareness of worldwide equities (intelligence and related products, personnel, facilities, products, equipment, and communications) in support of mission requirements and other statutory obligations.

4.2 **Cyber Capability Requirements Process:** The FBI seeks innovative approaches, methods, and technologies for data and requirements management, capability management, and budgeting to improve its enterprise cyber capability development and management processes.

4.3 **Enterprise Common Data Fabric Solutions:** The FBI seeks a common data fabric environment to provide law enforcement and intelligence communities with data and information-sharing capabilities that meet operational needs and mission objectives. Solutions must research and/or deliver:

- technologies and methods that enable information-sharing within the law enforcement and intelligence communities, or other government agencies, or
- frameworks or applications that support business processes, data access, and data correlation.

FBI particularly desires solutions that include the following functions/services:

- Common data fabric components
- Data governance and compliance logic
- Novel information/data sharing methods
- Applications supporting business processes
- Development within secure cloud networks
- System integration
- Application service technologies
- Mission service technologies
- Content management discovery and retrieval applications

5. Increase Organizational Effectiveness

5.1 **Performance Management:** The FBI seeks innovative approaches, methods, and technologies to build a Performance Management Program to be used to institutionalize a culture that encourages workforce behaviors that improve product and service delivery standards, increase transparency and accountability at all levels, and establish methods that use hard data and other criteria for meaningful assessment.

5.1.1 **Applicant Screening:** The FBI seeks to leverage innovative talent management approaches, methods, and technologies to optimize applicant screening/onboarding, career development, and advancement. Solutions must enable the FBI to ensure maximum productivity, quality of life, and achievement of career goals. Solutions must accurately derive applicant potential from biographical data, resumes, training history, leadership/management experience, and breadth/depth of professional expertise. Solutions must also provide hiring managers with a prioritized list of top applicants from a large pool, in accordance with customizable criteria. Solutions must integrate with existing FBI systems, rather than provide stand-alone software solutions. Solutions must be scoped as proofs-of-concept rather than large-scale systems.

5.1.2 **Career Advancement:** The FBI seeks a talent management system leveraging Machine Learning (ML) to manage, track, analyze, trend, and forecast employees' career advancement opportunities. Solutions must contain a web-based user interface for the FBI to enter and verify data, as well as relational tables with Human Resources (HR) data and employee training records. Solutions must allow talent managers to forecast and manage training courses, assignments, leadership tracks, promotion actions, Joint Duty Assignments (JDA), stretch assignments, and career moves that impact employees' overall development. Solutions must also house employee career counseling records, as well as a business analytics/visualization dashboard to quickly portray the health and welfare of the employee workforce.

5.1.3 **Workplace Learning and Performance:** The FBI seeks to expand its outreach and partnerships with divergent Research & Development (R&D) entities in order to anticipate developments in the human performance arena that impact workplace learning and performance practices.